
Quantum Computation and Communications (EE798V):

Problem Set 5

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Note: Assume all vector spaces to be finite-dimensional unless otherwise mentioned.

Q. 1: Let V be a complex vector space associated with an inner product and $W \subset V$ be an arbitrary subset. Recall that W is called a subspace of V if $\alpha \mathbf{w}_1 + \beta \mathbf{w}_2 \in W$, $\forall \mathbf{w}_1, \mathbf{w}_2 \in W$ and $\forall \alpha, \beta \in \mathbb{C}$.

- (a) Let $A \in \mathcal{L}(V)$ be an arbitrary linear operator and a one of its eigenvalues. Define the eigenspace $\mathcal{E}_a$:

$$\mathcal{E}_a = \{\mathbf{v} \in V \mid A\mathbf{v} = a\mathbf{v}\}$$

Show that $\mathcal{E}_a$ is a subspace of V .

- (b) Define the range of a linear operator $A \in \mathcal{L}(V)$ as

$$\text{range}(A) = \{A\mathbf{v} \mid \mathbf{v} \in V\}$$

Show that $\text{range}(A)$ is a subspace of V .

- (c) Let W be a subspace of V . Define the orthogonal space $W^\perp$ as:

$$W^\perp = \{\mathbf{v} \in V \mid \langle \mathbf{w} \mid \mathbf{v} \rangle = 0, \forall \mathbf{w} \in W\}$$

Show that $W^\perp$ is a subspace of V . Show that $W \cap W^\perp = \{\mathbf{0}\}$.

- (d) Let W_1 and W_2 be two subspaces of V . Define the direct sum as:

$$W_1 + W_2 = \{\mathbf{w}_1 + \mathbf{w}_2 \mid \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2\}$$

Show that $W_1 + W_2$ is a subspace of V . Additionally, show that

$$\dim(W_1 + W_2) \leq \dim(W_1) + \dim(W_2)$$

[Hint: Let $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ be a basis of W_1 and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be that of W_2 . Show that any element of $W_1 + W_2$ can be written as a linear combination of $\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n\}$.]

- (e) Recall that, if W_1, W_2 are orthogonal subspaces of V i.e., $\langle \mathbf{w}_1 \mid \mathbf{w}_2 \rangle = 0$, $\forall \mathbf{w}_1 \in W_1$, and $\forall \mathbf{w}_2 \in W_2$, then the direct sum $W_1 + W_2$ is written as the orthogonal sum $W_1 \oplus W_2$. Show that

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2)$$

- (f) Let W be a subspace of V and $W^\perp$ be its orthogonal space. Show that

$$V = W \oplus W^\perp$$

[Hint: Choose an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ of W such that $\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{u}_{m+1}, \dots, \mathbf{u}_n\}$ becomes an orthonormal basis of V . Use this to show $V \subset W \oplus W^\perp$. Prove the opposite inequality using exercise 1(d).]

- (g) Let $W_1 = \{(x, x) \mid x \in \mathbb{R}\}$ and $W_2 = \{(y, -y) \mid y \in \mathbb{R}\}$ be two subspaces of $\mathbb{R}^2$.

- (1) Show that W_1 and W_2 are orthogonal.
- (2) Show that $W_1 \oplus W_2 = \mathbb{R}^2$.
- (3) Verify that $\dim(W_1) + \dim(W_2) = \dim(W_1 \oplus W_2)$.

Q. 2: In the following, we use A to denote a normal operator associated with a Hilbert space, $\mathcal{H}$ i.e., $A \in \mathcal{L}(\mathcal{H})$ and $AA^\dagger = A^\dagger A$ where $A^\dagger$ is the adjoint operator of A i.e., $\langle \phi \mid A\psi \rangle = \langle A^\dagger \phi \mid \psi \rangle$ for any $|\phi\rangle, |\psi\rangle \in \mathcal{H}$.

- (a) Show that $A - aI$ is also normal where $a \in \mathbb{C}$ and I denotes the identity operator. In other words, show the following equality.

$$(A - aI)(A^\dagger - \bar{a}I) = (A^\dagger - \bar{a}I)(A - aI)$$

- (b) Assume that $|\psi\rangle \in \mathcal{H}$ is an eigenvector of A corresponding to the eigenvalue $a \in \mathbb{C}$. Using the above result, prove that $|\psi\rangle$ is also an eigenvector of $A^\dagger$ corresponding to the eigenvalue $\bar{a}$.
- (c) Let $\mathcal{E}_a$ be the eigenspace of A corresponding to the eigenvalue a . Prove that, if $a \neq b$, the eigenspaces $\mathcal{E}_a$ and $\mathcal{E}_b$ are orthogonal i.e., $\langle \phi | \psi \rangle = 0, \forall |\phi\rangle \in \mathcal{E}_a, \forall |\psi\rangle \in \mathcal{E}_b$. You might have to use exercise 2(b) to establish this result.
- (d) Let $\sigma(A)$ be the set of all eigenvalues of A . Due to the above result, one can define the following:

$$\mathcal{E} = \bigoplus_{a \in \sigma(A)} \mathcal{E}_a$$

Let $\mathcal{E}^\perp$ denote the space orthogonal to $\mathcal{E}$. Prove that, if $|\phi\rangle \in \mathcal{E}^\perp$, then $A|\phi\rangle \in \mathcal{E}^\perp$ i.e., $\mathcal{E}^\perp$ is closed under the operation A .

- (e) The above result allows us to define the restriction of A to $\mathcal{E}^\perp$. Let this restricted operator be denoted as A_0 . Argue that if $|\phi\rangle \in \mathcal{E}^\perp, |\phi\rangle \neq 0$ is an eigenvector of A_0 , it must also be an eigenvector of A .
- (f) Conclude that $\mathcal{E}^\perp$ must be trivial i.e., $\mathcal{E}^\perp = \{0\}$. Since $\mathcal{H} = \mathcal{E} \oplus \mathcal{E}^\perp$, we arrive at the following.

$$\bigoplus_{a \in \sigma(A)} \mathcal{E}_a = \mathcal{H}$$

- (g) Let $\{|\psi_{a,b}\rangle\}_{b \in \mathcal{J}_a}$ be an orthonormal basis (ONB) of $\mathcal{E}_a$. Using (f), prove that $\{|\psi_{a,b}\rangle\}_{a \in \sigma(A), b \in \mathcal{J}_a}$ is an ONB of $\mathcal{H}$.
- (h) Define the projection operator P_a as follows.

$$P_a = \sum_{b \in \mathcal{J}_a} |\psi_{a,b}\rangle \langle \psi_{a,b}|$$

Using (g), show the following completeness relation.

$$\sum_{a \in \sigma(A)} P_a = I$$

- (i) Prove the following equalities using the completeness relation stated above.

$$A = \sum_{a \in \sigma(A)} a P_a = \sum_{a \in \sigma(A)} \sum_{b \in \mathcal{J}_a} a |\psi_{a,b}\rangle \langle \psi_{a,b}|$$

The first expansion is typically known as spectral decomposition while the second one is known as eigen decomposition.

- (j) Consider a linear operator described by the following matrix.

$$A = \begin{pmatrix} 5 & 1 & 2 \\ 1 & 5 & -2 \\ 2 & -2 & 2 \end{pmatrix}$$

Answer the following questions regarding this operator.

- (1) Compute the eigenvalues of A .
 - (2) Obtain the projection operators corresponding to each eigenvalue.
 - (3) Check if the projection operators satisfy the completeness relation.
 - (4) Check if the spectral decomposition holds for A .
- (k) Repeat the same exercises for the following operator.

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Why are the answers to (3) and (4) different in this case?

Q. 3: In the following, $A, B \in \mathcal{L}(H)$ are commutative normal operators associated with Hilbert space, $\mathcal{H}$.

- (a) If $|\psi\rangle$ is an eigenvector of A corresponding to the eigenvalue $a \in \mathbb{C}$, show the following results.
- (1) $B|\psi\rangle$ is an eigenvector of A corresponding to the eigenvalue a .
 - (2) $B^\dagger|\psi\rangle$ is an eigenvector of A corresponding to the eigenvalue a .
- You might have to use exercise 2(b) to establish (2).

- (b) In summary, the above result indicates that the eigenspace $\mathcal{E}_a$ of a normal operator A is closed under another normal operator B , provided that $AB = BA$. In this case, we can define the restrictions of $B, B^\dagger$ to $\mathcal{E}_a$ as $B_a, B_a^\dagger$ respectively. Show that the operators $B_a B_a^\dagger$ and $B_a^\dagger B_a$ are well-defined i.e., every possible output of B_a is a valid input to $B_a^\dagger$, and vice versa. Additionally, show that B_a is normal i.e., $B_a B_a^\dagger = B_a^\dagger B_a$.
- (c) Using the above result and exercise 2(g), show that there exists an orthonormal basis $\{|\psi_{a,b}\rangle\}_{b \in \sigma(B_a)}$ of $\mathcal{E}_a$ such that each of its elements is a simultaneous eigenvector of A and B .
- (d) Using the above result, show that there exists an orthonormal basis $\{|\psi_{a,b}\rangle\}_{a \in \sigma(A), b \in \sigma(B_a)}$ of $\mathcal{H}$ such that each of its elements is a simultaneous eigenvector of A and B .
- (e) If $\{|\psi_i\rangle\}_{i=1}^d$ are simultaneous eigenvectors of A and B that form an ONB of $\mathcal{H}$, then using eigen decomposition, we can express A and B as follows.

$$A = \sum_{i=1}^d \lambda_i |\psi_i\rangle \langle \psi_i|, \text{ and } B = \sum_{i=1}^d \mu_i |\psi_i\rangle \langle \psi_i|$$

Verify that A, B are indeed normal and commutative. In conclusion, two operators $A, B \in \mathcal{L}(\mathcal{H})$ are simultaneously diagonalizable if and only if they are normal and commutative.

Q. 4: Prove the following properties.

- (a) Let $A \in \mathcal{L}(\mathcal{H})$ be a normal operator such that $\langle \psi | A \psi \rangle = 0, \forall |\psi\rangle \in \mathcal{H}$. Prove that $A = 0$.
- (b) Let $A \in \mathcal{L}(H)$ be an arbitrary linear operator. Show that $A - A^\dagger$ is normal.
- (c) Let $A \in \mathcal{L}(H)$ be arbitrary. Prove the following statements:
- (1) A is Hermitian if and only if $\langle \psi | A \psi \rangle$ is real $\forall |\psi\rangle \in \mathcal{H}$.
 - (2) A is Hermitian if and only if A is normal with real eigenvalues.

Recall that an operator $A \in \mathcal{L}(\mathcal{H})$ is called positive if $\langle \psi | A \psi \rangle \geq 0, \forall |\psi\rangle \in \mathcal{H}$.

- (d) Using exercise 4(c), conclude that any positive operator is necessarily Hermitian with non-negative eigenvalues.

Q. 5: Consider a qubit system whose state evolution is described by the following Schrodinger's equation:

$$i \frac{\partial}{\partial t} |\psi_t\rangle = \sigma_z |\psi_t\rangle$$

where σ_z is the Pauli Z matrix. Answer the following questions.

- (a) Write the spectral decomposition of σ_z .
- (b) Show that $|\psi_t\rangle = \exp(-i\sigma_z t) |\psi_0\rangle$ is a solution to the Schrodinger equation.
- (c) Express $U_t = \exp(-i\sigma_z t)$ in a 2×2 matrix form and verify that it is unitary.
- (d) Numerically compute the state vector at $t = 2$ provided that the initial state is $|\psi_0\rangle = \frac{1}{\sqrt{2}}(1, 1)^\top$.

Q. 6: Consider a normal operator $A \in \mathcal{L}(\mathcal{H})$ whose spectral decomposition is given as:

$$A = \sum_{a \in \sigma(A)} a P_a$$

- (a) Show the following equality via direct multiplication and the relation that $P_a P_b = P_a \delta_{a,b}$.

$$A^2 = \sum_{a \in \sigma(A)} a^2 P_a$$

- (b) Using the above result as a base case, establish the following relation via induction where m is an arbitrary positive integer.

$$A^m = \sum_{a \in \sigma(A)} a^m P_a$$

- (c) Use the above result to conclude that

$$f(A) = \sum_{a \in \sigma(A)} f(a) P_a$$

where f denotes a finite-degree polynomial. The above result is consistent with the definition of $f(A)$ where f is an arbitrary real-valued function.

- (d) Consider the normal operator A described in exercise 2(j). Compute $\sin(\pi A/36)$.
- (e) Using the above result, obtain $\mathbf{E}_\psi[\sin(\pi A/36)]$, and $\text{Var}_\psi[\sin(\pi A/36)]$, where $|\psi\rangle = \frac{1}{\sqrt{3}}(1, 1, 1)^\top$.

Q. 7: Assume that the state of a qubit system is $|\psi\rangle = \frac{1}{\sqrt{5}}(1, 2)^\top$, and an observer is attempting to measure two physical quantities described by the Hermitian operators σ_x and σ_z .

- (a) Compute the standard deviation $\Delta_\psi[\sigma_x]$.
- (b) Compute the standard deviation $\Delta_\psi[\sigma_z]$.
- (c) Compute the expectation $\mathbf{E}_\psi([\sigma_x, \sigma_z])$.
- (d) Check if the Heisenberg-Robertson uncertainty principle holds for the above observables.

Q. 8: Consider a 3-level quantum system with $\mathbb{C}^3$ as its associated Hilbert space. Assume that the state of the system is $|\psi\rangle = \frac{1}{\sqrt{2}}(1, 1, 0)^\top$. An observer is measuring two physical quantities described by the following Hermitian operators.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 2i \\ 0 & -2i & 1 \end{pmatrix}$$

- (a) Show that A and B commute.
- (b) Find the eigenvalues of A and B , and a set of simultaneous orthonormal eigenvectors.
- (c) Using the above result, develop a measurement scheme that simultaneously generates measurement values of A and B .
- (d) Calculate the joint probabilities $\Pr(A = a, B = b)$ where $a \in \sigma(A)$ and $b \in \sigma(B)$.
- (e) Verify that the marginal probabilities obtained from the above result are consistent with the Born rule. This indicates that A and B are simultaneously measurable.